Deployment Headline: Cross-Country Fiscal-Shock Inventory

Full C1 → C0 → C2 deployment pipeline, by country

Esteban Degetau
World Bank
estebandegetau@gmail.com

Agustín Samano
World Bank
asamanopenaloza@worldbank.org

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Framing

This is the deployment headline: the end-to-end output of the validated **C1 → C0 → C2** pipeline run on each deployment country's full document corpus. It presents the same figures and tables as the Malaysia EN/BM consistency report, but with the comparison axis swapped from **language** to **country**.

The pipeline runs independently per country. C1 identifies fiscal measures, C0 aggregates measure names into acts, and C2 classifies each act's motivation, sign, and exogeneity. Today only Malaysia is deployed, so every comparison figure shows a single country; the notebook auto-extends to a panel per country as Indonesia, Thailand, the Philippines, and Vietnam are onboarded.

Each country's acts are clustered within that country only. We do **not** attempt to merge acts across countries: a Malaysian GST and a future Indonesian GST are distinct national measures, so there is no cross-country analog of the consistency report's joint EN/BM merge probe. The per-country C0 cluster summary below takes that slot.

This is a deployment inventory, not a validity test. There are no ground-truth labels for any deployment country; the act-level outputs are inputs requiring expert validation (Phase 2 success criteria: ≥80% expert agreement on C1, ≥70% on C2), not research findings.

Scope

Figure 1 shows each country's corpus extent: pages and chunks per document year, summed across that year's documents. As more countries are added, comparable totals indicate comparable corpus depth; large gaps flag coverage asymmetries that bound everything downstream.

Figure 1: **Corpus scope.** Pages and chunks per country and document year, summed across that year's documents.

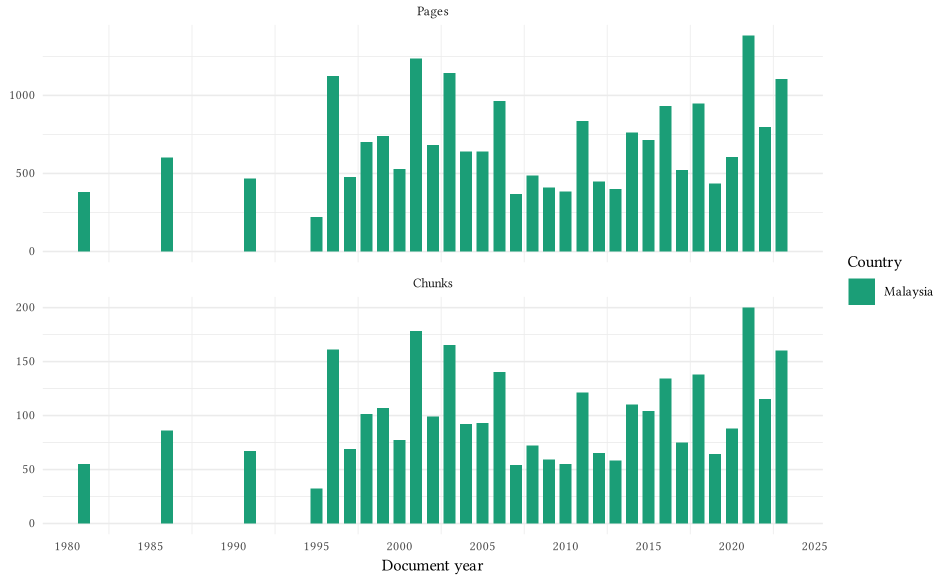


Table 1 reports the headline tally per country: the document-year span analysed, the corpus size (documents, pages, chunks), the fiscal measures C1 surfaces, the acts C0 clusters them into, and how many of those acts C2 classifies as exogenous.

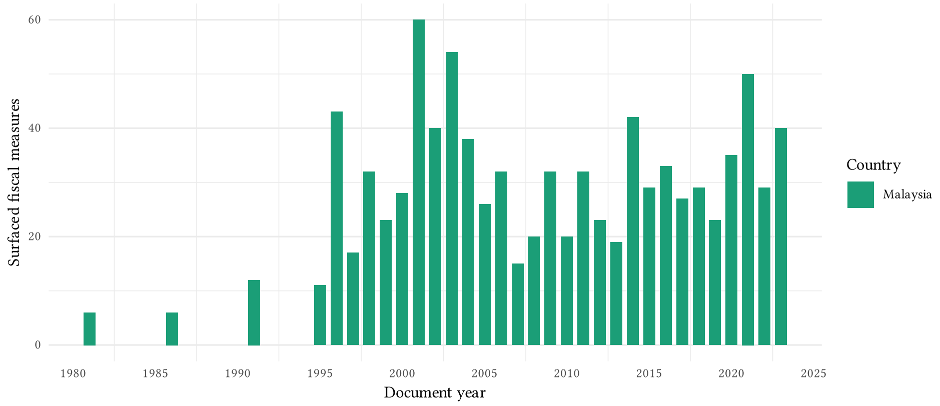
Table 1: Per-country headline tally: corpus size, surfaced and clustered fiscal measures, and exogenous acts.

Country	Years analysed	Documents	Pages	Chunks	Surfaced fiscal measures	Clustered fiscal measures	Exogenous fiscal measures
Malaysia	1981–2023	113	22050	3194	877	805	603

C1-step comparability

Before aggregation, what set of fiscal measures does C1 surface for each country? This is where real extraction differences live, independent of how acts are later aggregated. Figure 2 counts the distinct fiscal measures C1 surfaces per year, by country.

Figure 2: **Surfaced fiscal measures per year, by country (pre-aggregation).** The extraction load that bounds any downstream aggregation.



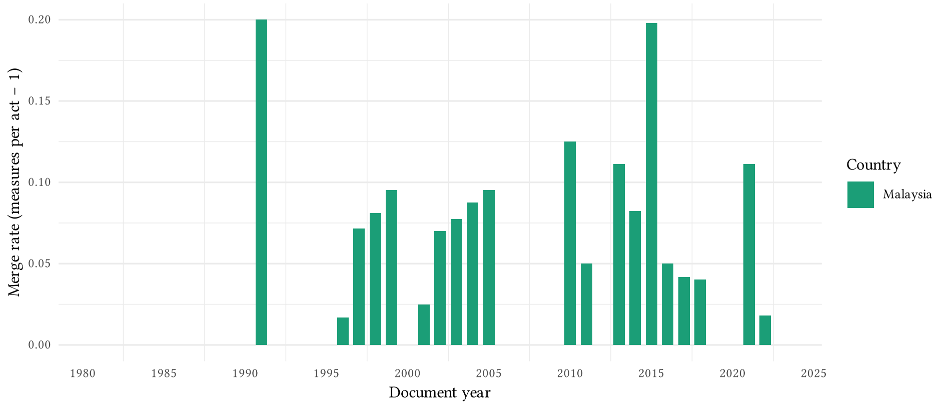
C0 aggregation diagnostics

C0 collapses C1 measure names into act-level clusters. The figures below describe that aggregation per country.

Merge rate

Figure 3 shows the merge rate, (surfaced measures / distinct acts) − 1, by year and country, averaged across the year's documents. Zero means no merging (every surface form is its own act); higher means more variants absorbed per act.

Figure 3: **C0 merge rate (surfaced measures per act − 1) per year and country.** Higher bars mean more measure variants absorbed into each canonical act.



Per-country cluster summary

How many distinct acts does C0 produce for each country, and how many measure variants does each act absorb on average? Table 2 is the per-country analog of the consistency report's joint merge-rate probe.

Table 2: Per-country C0 cluster summary: distinct acts, measure variants, and cluster sizes

Country	Measure variants	Acts (clusters)	Merge rate	N singletons	Largest cluster
Malaysia	877	805	0.09	755	7

Figure 4: **Acts by their *enacted* status.** An act is said to be enacted if C2b found sufficient evidence in that regard.

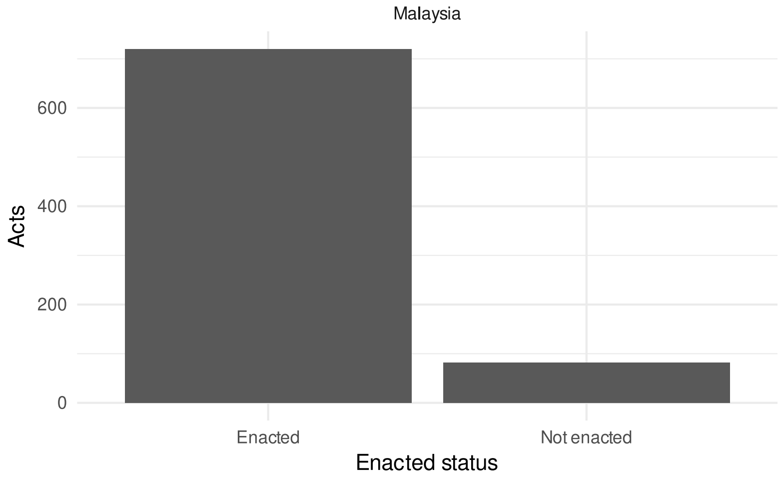
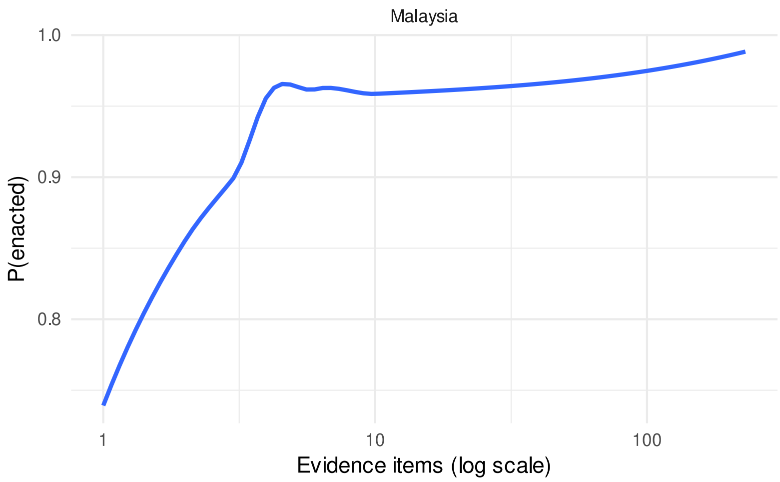


Figure 5: Probability of enacted by amount of evidence



Identification by frequency

Notice that the language used by Romer and Romer (n.d.) to define what constitutes as a "significant change to fiscal liabilities" over-identifies seemingly innocuous fiscal changes when operationalizing it through the LLM chain we developed. We showed whilst developing C0 that we can recover Romer and Romer (n.d.) 's taxonomy of US fiscal changes by selecting a threshold of the most discussed acts in the corpus.

This section asks what is a sensible threshold for selecting meaningful fiscal changes that surfaced from our pipeline, that we know are over-identified. While RR's taxonomy included 49 distinct fiscal changes spanning 58 years (from 1945 to 2003), the acts aggregated by C0 number 805 over 42 years (from 1981 to 2023). Thus we say RR's language, as operationalized by our pipeline, over-identifies fiscal changes.

To sift through the most relevant changes, without yet knowing the fiscal magnitude of each, we recur to a frequentist approach. We use two raw variables to map on to a relevance index. First, more relevant acts should have produced more **evidence items**, as surfaced by C2a. Moreover, more relevant acts should also be discussed in **more parts of the corpus**. So, we will want the acts for which we have a lot of evidence to extract their motivation from, and that were more widely discussed in the government documents.

Notice in Figure 6 that, some acts have just one piece of evidence extracted from one chunk, in all likelihood, these acts are false positives (e.g. insignificant acts that RR would not have identified as significant enough); these acts should score the lowest with our relevance measure. On the other extreme, a single act surfaced some 250 evidence items from almost 50 different chunks; this act is most certainly important enough that RR would have identified it, so it should get the highest available score. Consider the following Cobb-Douglass relevance function

$$\text{relevance} = \text{evidence items} \cdot \text{chunks addressed}$$

Then, how is *relevance* accumulated by act? Figure 7 shows that 16 acts accumulate 80% of the relevance, as defined above.

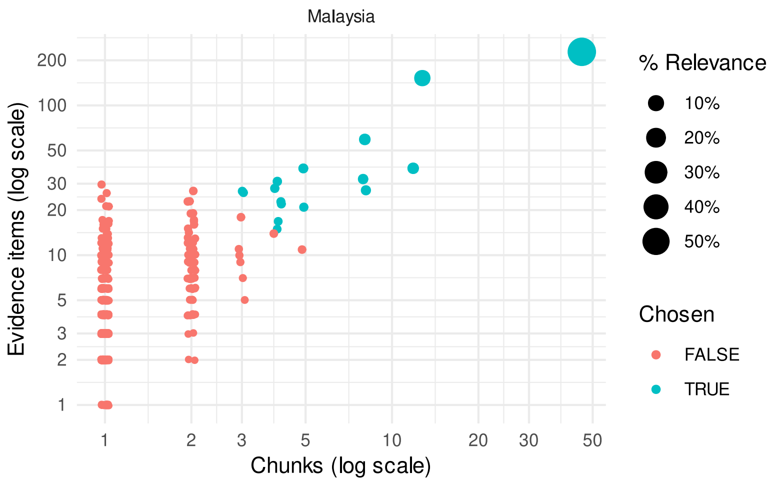


Figure 6: **A measure of relevance**. Relevance as a mapping of evidence items and chunks were the act was found.

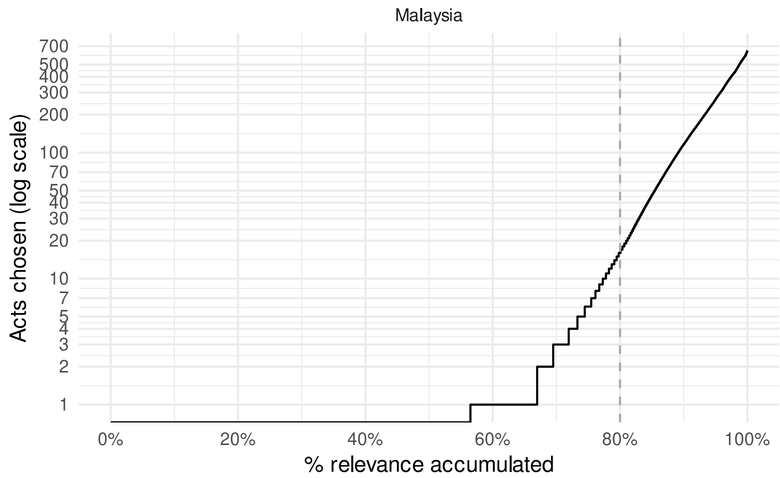


Figure 7: **Choosing acts according to accumulated relevance**. The dashed line marks the 80% relevance cut

Figure 8: **Timeline of un/chosen acts**.

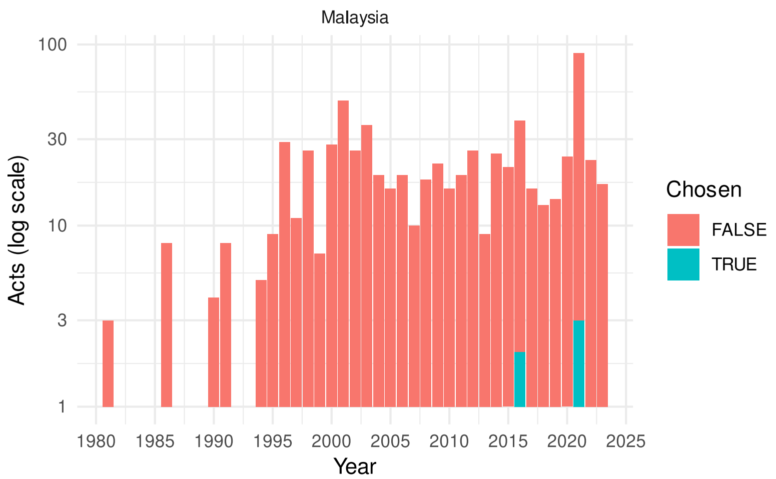


Table 3 shows the chosen acts. For example, the **Goods and Services Tax (GST)** pools 230 evidence items drawn from 46 chunks spanning 7 EN/BM surface forms that C0 merged into one act; C2b read that consolidated evidence and classified the act's motivation as deficit-driven (revenue increase).

Table 3: Acts that accumulate 80% of relevance.

Act name	Year	Motivation	Effect	Evidence items	Chunks	Accumulated relevance
Goods and Services Tax (GST)	2015	Deficit-driven	+	230	46	56.5%
National Economic Recovery Plan (NERP)	1998	Counter-cyclical	+	151	13	67.0%
Pakej Perlindungan Rakyat dan Pemulihan Ekonomi (PEMULIH)	2021	Counter-cyclical	+	59	8	69.5%
Akta Penggalakan Pelaburan 1986	1986	Long-run	-	38	12	71.9%
Anti-Money Laundering Act 2001	2001	Long-run	+	32	8	73.3%
Bantuan Rakyat 1Malaysia (BR1M)	2016	Counter-cyclical	+	27	8	74.5%
Prihatin Rakyat Economic Stimulus Package (PRIHATIN)	2021	Counter-cyclical	+	38	5	75.5%
PRIHATIN Package	2020	Counter-cyclical	+	31	4	76.1%
Pakej	2021	Counter-	+	28	4	76.7%

Year: the year parsed from the act name when it contains one, otherwise the median year of the source documents that discuss the act.

Evidence items: motivation quote-and-signal pairs C2a extracted from the documents, pooled across all of the act's chunks.

Act inventory marginals

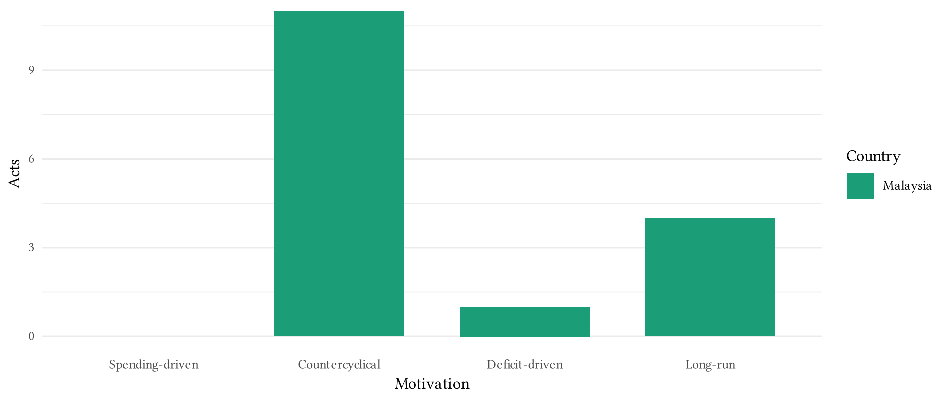
The marginals and the timeline below are restricted to the chosen acts (the 80%-relevance cut defined above), not the full over-identified inventory.

Pooling all of a country's documents, Table 4 reports the act count and the exogenous/endogenous split, and Figure 9 shows the motivation-label marginal distribution.

Table 4: Per-country act inventory: counts and exogeneity split

Country	C0 acts	Exogenous	Endogenous
Malaysia	16	5	11

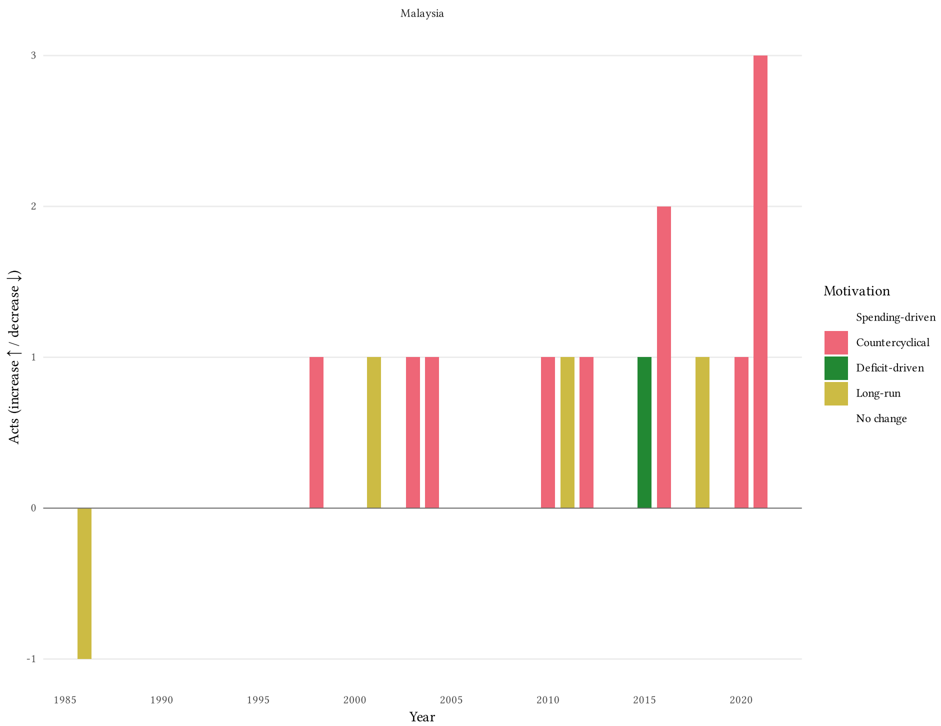
Figure 9: **Motivation-label marginal distribution by country.** How many acts each country assigns to each motivation category.



Headline: act inventory timeline

The pipeline produces, per country, a set of acts each carrying a C2 motivation label, sign, and exogeneity. The headline is a timeline of the **chosen** acts: for each year and country, acts are counted into motivation × sign cells and drawn as a diverging stacked bar. Increases grow the bar upward, decreases downward, no-change acts sit at the baseline, colour encodes motivation, faceted by country. Acts are dated by the single act-level year used throughout this section (parsed from the act name when present, else the modal source-document year).

Figure 10: **Chosen-act counts by year, by country.** Diverging stacked bars: increases grow upward, decreases downward, no-change acts sit at the baseline; colour = motivation.



Interpretation

Read the evidence in pipeline order:

- **C1 step.** The comparability figure describes what raw material the extractor surfaces per country. Any extraction gap here bounds everything downstream and is not fixable by the aggregator.
- **C0 step.** The merge-rate figure and cluster summary describe how aggressively C0 collapses measure variants into acts. A country whose corpus repeats the same handful of acts across many documents merges more.
- **Identification by frequency.** R&R's language over-identifies through the LLM chain, so we keep the acts that accumulate 80% of relevance (evidence items × chunks). Everything from the marginals onward is restricted to that chosen set.
- **C2 / headline.** The timeline is the deliverable: each country's chosen acts placed by year, direction, and motivation. It is the starting point for expert review, not a finding.

Implications

The same pipeline, codebooks, and figures apply to every country: onboarding the next one adds a corpus and a configuration entry, and this notebook gains a panel. The act inventory each country produces is the artifact an expert reviews against the Phase 2 agreement targets; comparable structure across countries (similar compression, plausible motivation marginals, no degenerate years) is the signal that the country-agnostic codebooks transferred.

Romer, Christina D., and David H. Romer. n.d. *A Narrative Analysis of Postwar Tax Changes.*